

1. for any positive integer, there exists a second positive the square of which is equal to the first integer.

x = Any positive integer ($x > 0$), y = Any positive ($y > 0$), $P(x, y)$ is $y^2 = x$

$\forall x \exists y P(x, y)$

- Because y can be any positive, not just an integer then for there to be at least one y value for every x value so that $y^2 = x$, y would have to be the square root of x .

$$y^2 = x$$

$$\sqrt{y^2} = \sqrt{x}$$

$$y = \sqrt{x}$$

- If y is the square root of x , even if it is not an integer then there will be one y value in which $y^2 = x$. ■

2. for any positive integer, there exists a second positive integer which is greater or equal to the square of the first integer.

x = Any positive integer ($x > 0$), y = Any positive integer ($y > 0$), $P(x, y)$ is $y \geq x^2$

$\forall x \exists y P(x, y)$

- Because y must be a positive integer then at least one y value will exist for every x value where $y \geq x^2$ because x must always be an integer, and any integer squared yields another integer.
- For each value of x , there is a y so that $y = x^2$ so there is at least one value for every x value so that $y \geq x^2$ is true. ■

3. for any positive integer, there exists a second positive which is greater or equal to the square of the first integer, but smaller than the cube of the first integer.

x = Any positive integer ($x > 0$), y = Any positive ($y > 0$), $P(x, y)$ is $x^3 > y \geq x^2$

$\forall x \exists y P(x, y)$

- For each value of x , there is a y so that $y = x^2$ so there is at least one value for every x value so that $y \geq x^2$ is true proving the second half of the $x^3 > y \geq x^2$ propositional to be true.
- If there is a y value where $y \geq x^2$ such that $y = x^2$, then the propositional $x^3 > y \geq x^2$ can have its y value replaced such that $x^3 > x^2 \geq x^2$ and thus $x^3 > x^2$ which is a true statement.

- This means that for all values of x , there exists at least one y value such that it is greater or equal to the square of x , but smaller than the cube of x . This is at least where $y = x^2$ for each value of x . ■